American Economic Journal: Microeconomics 2022, 14(4): 616–644
https://doi.org/10.1257/mic.20200311

Making Elections Work: Accountability with Selection and Control†

By Vincent Anesi and Peter Buisseret*

We study the limits of dynamic electoral accountability when voters are uncertain about politicians' characteristics (adverse selection) and their actions (moral hazard). Existing work argues that voters cannot achieve their first-best payoff. This is attributed to inherent deficiencies of the electoral contract, including voters' inability to precommit, and the restriction to a binary retention-replacement decision. We provide conditions under which voters can, despite these constraints, obtain arbitrarily close to the first-best payoff in an equilibrium of the electoral interaction. Our paper resolves that there need not be a trade-off between selection and control. (*JEL* D72, D82)

In a representative democracy, elections enforce the accountability of politicians to citizens. However, elections suffer from obvious limitations. They offer voters only a single instrument to achieve two distinct goals: selecting better politicians and controlling their behavior in office. They pit uninformed voters against politicians that have superior knowledge about the policymaking context, their actions in office, and their own characteristics. Elections are also coarse contracts: they restrict voters to a binary decision to replace or retain an incumbent. And voters cannot precommit to these decisions: their choices must be optimal at the time they are made.

In view of these limitations, scholars argue that electoral institutions themselves are a barrier to political accountability, even setting aside issues of voter rationality. Some worry that a lack of commitment precludes the use of elections to select competent politicians and control their behavior (Fearon 1999; Besley 2006); on the contrary, there is a *trade-off* between these objectives, so that voters may "strive for one goal only at the expense of the other" (Manin, Przeworski, and Stokes 1999, 47).[1]

* Anesi: Department of Economics and Management, University of Luxembourg (email: vincent.anesi@uni.lu); Buisseret: Department of Government, Harvard University (email: pbuisseret@fas.harvard.edu). Leeat Yariv was coeditor for this article. We are grateful to Benoit Crutzen, John Duggan, Alex Fouirnaies, Navin Kartik, Antoine Loeper, Mikhail Safronov, Daniel Seidmann, Cecilia Testa, and Richard Van Weelden for helpful comments and suggestions.

† Go to https://doi.org/10.1257/mic.20200311 to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

[1] The argument is as follows: if incumbents and challengers are homogeneous (as in Ferejohn 1986 and subsequent work in that tradition—recently, Zudenkova 2018), voters are indifferent between retention and replacement. They can therefore commit to sanction poor incumbent performance and, thus, *control* politicians' incentives. If, however, politicians are heterogenous in a way that matters for future payoffs—for example, if they are distinguished

Others argue that the coarseness of the voter's binary decision to replace or retain the incumbent is the decisive limitation, and "express doubts that voters, even rational and well-informed voters, could exercise control over the elected official with such a blunt tool" (Ingham 2019, 129).

These issues are explored in principal-agent models of electoral accountability. While the vast majority of theoretical work focuses on two-period interactions, pessimistic conclusions also arise in models with many periods. Even for arbitrary degrees of voter patience (discount factors), equilibria of the dynamic interaction between voters and politicians hitherto retain the trade-off between selection and control (Banks and Sundaram 1993) and yield "an inherent bound on responsive democracy" (Duggan and Martinelli 2017, 957).

Our paper revisits these concerns in a dynamic model of electoral accountability with both adverse selection and moral hazard. We ask: Is the trade-off between selection and control fundamental to the electoral mechanism? Are the deficiencies of the electoral mechanism to blame for an upper bound on political accountability? What *is* the upper bound on political accountability? As Gailmard (2010, 93) puts it, "even in a world of perfect voter attention and rationality, do electoral institutions themselves limit the accountability of policy to voters' interests?"

We provide a positive and remarkable answer: if voters do not discount future policy outcomes too much, they can closely approximate the first best; i.e., they can obtain the payoff that they would receive if they were perfectly informed about both politicians' actions and their characteristics. Specifically, we show the following result. Suppose that voters could implement the first-best payoff in an idealized setting with full commitment and monetary transfers. Then, there exists an equilibrium of the dynamic electoral interaction—in which voters have *neither* commitment *nor* transfers—that approximates this payoff arbitrarily closely. Our result stands in sharp contrast to existing results in the literature on dynamic elections: it resolves that there need not be a trade-off between selection and control even in the absence of commitment and that coarseness of the electoral mechanism need not impede accountability.

Our Approach.—Our framework studies a long-run (infinite horizon) relationship between a set of political candidates and a representative voter. In each period, an election is held, and the voter derives a stage payoff from a policy outcome generated by the elected policymaker. This stage payoff depends stochastically on the officeholder's type (i.e., her ability) and the action that she takes in office (i.e., her effort). The voter observes neither politicians' types nor their actions: her only sources of information are cheap talk messages that candidates may send during each election campaign and the policy outcome at each date. Politicians' types may vary over time.

The first-best outcome is that the voter appoints the best candidate (*selection*), who subsequently exerts maximal effort (*control*) in every period. We initially

by expected ability—then voters are not indifferent and prefer to use their electoral decisions to maximize the quality of *selection*. But this removes their flexibility to control politicians by making commitments to sanction and reward past behavior.

consider an idealized setting in which the voter can commit to her reelection decision and a monetary reward scheme, as a function of candidates' announcements and her policy outcomes. In that context, we say that the first best is *implementable with transfers* if there exists a reward scheme that incentivizes each candidate to reveal her type and choose maximal effort if elected.

We transition to an infinite horizon setting in which the voter's only instrument for selection and control is her decision to reelect or replace the incumbent. Naturally, we do not allow for any binding commitments. We adopt the classical mechanism design approach to dynamic games with imperfect information by using a candidate's future payoffs to simulate the transfers that she would receive in our hypothetical static benchmark. The idea underlying this approach is straightforward: if the variation in a candidate's future payoffs from today's choices replicates how transfers vary with her choices in the static interaction, her incentives to reveal her type and work hard in the static context with transfers can also be replicated in a perfect Bayesian equilibrium (PBE) of the dynamic environment *without* transfers. Our main result establishes that this can be achieved.

Our result does not merely show that it is possible to achieve selection and control in an equilibrium: we also explicitly characterize the voter's decision rule, which has both intuitive and substantively interesting properties.

First, the rule is "retrospective" in a strong sense: it depends on not only the current incumbent's performance but also the past performances and campaigns of *all* candidates. Like Gordon, Huber, and Landa (2007), we therefore emphasize the importance of electoral challengers for outcomes, even when they do not win. Second, the voter chooses the lowest-quality candidate at each election with an arbitrarily small probability, which increases as the past performance of the candidates deteriorates or when the voter believes that candidates' announcements during previous campaigns were lies. Thus, it is part of an optimal strategy to elect on rare occasions (and only temporarily) a bad candidate when the voter feels that she has been repeatedly let down by the remaining candidates (e.g., Di Tella and Rotemberg 2018). Third, when the voter decides that a candidate's campaigns are no longer credible—which occurs with positive probability in equilibrium—she disregards the candidate's messages for a number of subsequent elections. During those elections, she "tunes out" and simply ignores that candidate.

Are there any contexts in which the first best can be implemented with transfers? A special case of our policy environment is the widely used *linear outcome* model, in which the voter's payoff is a linear function of the incumbent's type, her effort, and a noise term (popularized by the textbook presentation of Persson and Tabellini 2002). We show that in the linear outcome model, the first best is implementable with transfers. Our result immediately yields that if agents do not discount future policy outcomes too much, then there exists an equilibrium of the dynamic game in which the first-best payoff is arbitrarily well approximated. Further, the contrapositive of our result implies that if selection and control cannot be achieved in an equilibrium when agents are sufficiently forward-looking, it is *not* because the electoral mechanism is coarse and voters lack commitment. Rather, it is because they could not achieve the first best even if they had commitment and transfers.

I. Related Work

Our theoretical result does *not* dismiss concerns about the practical limits of political accountability. Concerns about these practical limits focus on not only institutions but also limits to voter rationality or farsightedness. However, what can be achieved in practice is bounded from above by what can be achieved in principle. And what can be achieved in principle is a theoretical question that remains "one of the central conceptual issues of concern in the literature on the political economy of delegation" (Landa 2010, 1). Our goal is to address the open question of what limitations are *necessary* and the inevitable consequences of the limitations of the electoral mechanism, quite apart from the limits of voter competence. Is it true, as Manin, Przeworski, and Stokes (1999, 50) claim, that "citizens' control over politicians is at best highly imperfect" and that the fault lies in the electoral mechanism? Our result suggests that the answer is *no*: the failure of accountability derives not from intrinsic limitations of the mechanism but from how it is used in practice.

We describe the four features of our framework that are necessary for our result and their relationship to existing work.

(i) We study an infinite horizon with long-lived voters who internalize their future payoffs (as in, for example, Banks and Sundaram 1993, 1998; Duggan 2014, 2017). These voters can commit (in equilibrium) to punish deviating candidates, because any failure to do so generates a costly norm of unaccountability.[2]

(ii) A candidate who loses today's election can be elected again in the future, and our solution concept allows for history-dependent strategies. While the vast majority of existing work imposes that losing incumbents cannot subsequently return to office (Ferejohn 1986 is a notable exception), in reality politicians *do* serve nonconsecutive terms both in the United States[3] and elsewhere (Roche 2020).[4]

The combination of (i) history dependence and (ii) allowing politicians who lose an election today to hold office in the future allows us to fully exploit the power of repeated interactions. Many existing models of dynamic electoral accountability limit the history dependence of agents' strategies explicitly, or implicitly by way of short-lived voters (e.g., Kartik and Van Weelden 2019)[5] or two-period term limits (Banks and Sundaram 1998; Duggan 2017). Acharya, Lipnowski, and Ramos (2020) study nonstationary equilibria in a model of pure moral hazard. Finally, Duggan and Martinelli (2017) examine a class of stationary equilibria for an infinite horizon model with both adverse selection and moral hazard. In sharp contrast to

[2] Specifically: whenever the voter deviates from the prescribed selection rule, she is punished by perpetual play of the babbling equilibrium.

[3] In the House of Representatives, David Price (NC-04), Steve Chabot (OH-01), Mike Fitpatrick (PA-08), Tim Walberg (MI-07), Anna Kirkpatrick (AZ-02), and Bob Dold (IL-10) all lost and subsequently reclaimed their seats. In New Hampshire's first congressional district, incumbent Carol Shea-Porter lost her seat to Frank Guinta in 2010 and won the seat back from him in 2012, only to lose the seat to Guinta again in 2016. The US Senate also offers many examples: see https://centerforpolitics.org/crystalball/senatenonconsecutiveterms/.

[4] In a recent survey, Duggan and Martinelli (2017, 957) argue that the canonical framework "should be extended to incorporate a meaningful model of political careers, including … the possibility that a former office holder reenters the political scene (rather than the current standard of a random draw without replacement)."

[5] Kartik and Van Weelden (2019) extend their term-limited benchmark to a setting with no term limits but impose that after a second period in office, a politician's (persistent) type is exogenously revealed.

our result, they find that, assuming such equilibria exist,[6] the voter's commitment problem implies that her equilibrium payoffs are bounded above for arbitrary discount factors, a "negative result [that] establishes an inherent bound on responsive democracy."

(iii) Candidates can communicate their private information during electoral campaigns by way of cheap talk messages. This is the main instrument for voters to select the best candidates rather than relying on past performance, and it is a novel feature of our framework. We emphasize that the campaign announcement is pure cheap talk; in particular, it is *not* a commitment to take any action either if that candidate is elected today or at any future date.

(iv) Voters can observe an incumbent's policy outcome before the next election. This assumption is common to many models of electoral accountability (e.g., Duggan 2017 and Duggan and Martinelli 2017).

In our model, voters' stage payoffs depend directly on an incumbent's type and action. Our general formulation specializes to environments that are widely used in applied work. This includes contexts in which ability and effort enter voter stage payoffs as complements (e.g., the product of type and effort as in Ashworth and Bueno de Mesquita 2006; Alt, Bueno de Mesquita, and Rose 2011; Landa and Le Bihan 2018; Fouirnaies and Hall 2018; Camargo and Degan 2019) or substitutes (such as the linear outcome model, used in Persson and Tabellini 2002; Ashworth 2005; Gehlbach 2006; Alesina and Tabellini 2007, 2008; Ashworth and Bueno de Mesquita 2017).[7]

Some papers consider the case in which a politician's type affects her (marginal) cost of actions in both two-period settings (Duggan and Martinelli 2020) and infinite horizon settings with two-period term limits (Banks and Sundaram 1998; Duggan 2017). Banks and Sundaram (1993) construct equilibria in trigger strategies in an infinite horizon setting without term limits. Whenever output falls below a threshold, the incumbent shirks at all future periods and is therefore removed. This construction depends sensitively on the assumption that a politician's type only affects that politician's payoff, since it renders the voter indifferent over all shirking politicians, regardless of type. It also exhibits the classical tension between selection and control: the trigger strategy requires voters to remove a politician that is very likely to be a good type when output is low.

Our approximation of the voter's first-best payoff in this paper may bring to mind the game-theoretical literature on folk theorems, which are also based on approximating payoff vectors in the players' Pareto frontier. At a substantive level, our different approach permits us to study not only agents' payoffs but also their equilibrium behavior. We view this as a critical part of our contribution. At a more technical level, our model has distinctive features that do not allow us to apply standard results from the folk theorem literature. Namely: (i) our game exhibits both adverse selection and moral hazard; (ii) it has interdependent values, since the voter's utility depends on the winning candidate's type; (iii) types are serially correlated; and

[6] There are presently no general results on the existence of stationary equilibrium in these environments.

[7] Though we interpret types as reflecting candidates' abilities—ranked higher or lower independent of policy—our result extends to more general (possibly policy-dependent) orderings.

(iv) in each period, the public signal (output) is generated not only by the officeholder's hidden action but also by her type.

Our framework can extend to settings where politicians' actions are observable but where those actions are taken after politicians learn some private information that matters to voters. For example, the policy outcome might depend on a state of the world, and the politician privately observes a noisy signal of the state (as in Canes-Wrone, Herron, and Shotts 2001). One can translate this additional source of incomplete information into a moral hazard variable by defining the incumbent's actions as mappings from her set of signals into her set of moves, thus obtaining unobservable actions as in our model.

II. Model

Agents.—We consider an interaction between a representative voter and a finite set $C \equiv \{1, \ldots, n\}$ of candidates and a single low-quality candidate, denoted by 0. The interaction unfolds over an infinite number of discrete periods $t = 1, 2, \ldots$.

Adverse Selection.—In each period $t \in \mathbb{N}$, each candidate i has a type $\theta_i^t \in \Theta_i$. This type represents the candidate's quality or competence. The candidates' type sets are finite, disjoint subsets of $\mathbb{R}$. These types are distributed according to independent, irreducible Markov chains on $\Theta = \Theta_1 \times \cdots \times \Theta_n$.[8]

Moral Hazard.—In each period t, the elected candidate, say i, chooses an unobservable action a_i^t from a finite set $A \subset \mathbb{R}_+$ that includes 0. This action can be interpreted as the level of effort exerted in office. We let a^* denote the maximal effort level, i.e., $a^* \equiv \max A$. The officeholder's action stochastically generates an observed policy outcome $y^t \in [\underline{y}, \bar{y}]$. The cumulative distribution function governing y^t, given the officeholder's type θ_i^t and action a_i^t, is denoted by $F(\cdot \mid \theta_i^t, a_i^t)$ and has full support on $[\underline{y}, \bar{y}]$. We assume that for all $y \in [\underline{y}, \bar{y}]$, $F(\cdot \mid \theta_i^t, a_i^t)$ decreases in θ_i^t and a_i^t and strictly decreases in at least one argument. The voter observes only the realized policy outcome.

Timing.—Events unfold as follows in each period t:

(i) Each candidate privately learns her type.

(ii) Each candidate simultaneously makes a public announcement from a set of messages M_i.

(iii) The voter elects a candidate from $\{0, 1, \ldots, n\}$.

(iv) The elected candidate i selects action a_i^t from A.

[8] Although this formulation precludes fully persistent types, we can approximate full persistence by assuming that types vary with arbitrarily small probabilities.

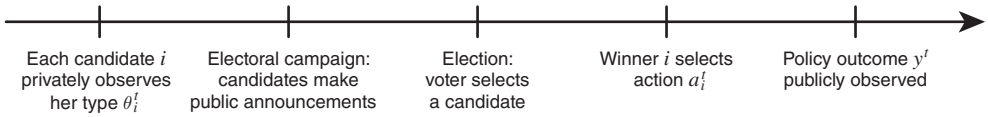

FIGURE 1. TIMING IN EACH PERIOD $t = 1, 2, \ldots$

(v) All agents observe the policy outcome y^t.

This timing is highlighted in Figure 1.

We denote the voter's (possibly random) decision by $\pi = (\pi_1, \ldots, \pi_n) \in [0,1]^n$, such that $\sum_{i=1}^n \pi_i \leq 1$, where π_i is candidate $i \in C$'s probability of winning the election. Thus, the low-quality candidate's probability of winning is $1 - \sum_{i=1}^n \pi_i$.

Payoffs.—The voter's stage game payoff is given by $v(y^t)$, where $v: \mathbb{R} \to \mathbb{R}$ is a continuous, strictly increasing function; the elected candidate's stage game payoff is given by $B - c(a^t)$, where $B > 0$ is an office rent and $c: \mathbb{R}_+ \to \mathbb{R}$ is a strictly convex cost function with $c(0) = 0 < c'(0)$; and all the candidates who lose the election at date t receive a payoff of 0 at that date. Time is discounted at a common rate $\delta \in [0,1)$. Agents maximize their average discounted utilities.

Equilibrium.—We study perfect Bayesian equilibria, allowing agents to play mixed strategies.

Additional Assumptions.—We impose additional assumptions that ease the analysis. First, we assume that candidate 0 has a single type θ_0 and that $\theta_0 < \min_i(\min \Theta_i)$. In other words, the quality of the candidates in C is always higher than that of the "bad" candidate 0. This assumption could be replaced by a weaker restriction that the probability with which this candidate is the very best candidate at any date is not too large. Second, we assume that the set of messages M_i is sufficiently rich, in the sense that $\Theta_i \times [0,1]$ is embedded in M_i. This allows us to generate a public randomization device that makes deviations by the voter detectable. Third—and to avoid trivialities—we assume that the office rent outweighs a candidate's costs from choosing maximal effort, i.e., $B > c(a^*)$, and we assume that for every candidate i, there is a positive probability at each date of a profile θ such that candidate i is the most able candidate, i.e., $\theta_i = \max\{\theta_j : j = 1, \ldots, n\}$.

Discussion.—Our candidates care only about winning office and the cost of any effort that they expend. However, we could assume that candidates are also policy motivated. This would strengthen the voter's ability to control politicians, by increasing the number of dimensions along which she may construct incentives—as in Bernhardt, Câmara, and Squintani (2011) or Van Weelden (2013).

It also worth remarking that our assumption that an incumbent takes a single action $a \in \mathbb{R}$ is only for expositional convenience. All the arguments and results in this setting can easily be translated (at the cost of heavier notation) to one where incumbents choose action *vectors* $a = (a_1, \ldots, a_m) \in \mathbb{R}^m$, $m \in \mathbb{N}$, which in turn

generate multidimensional output vectors $y = (y_1, \ldots, y_d)$, $d \in \mathbb{N}$. Thus, our analysis also speaks to a related substantive concern about the limits of elections: they endow voters with a *single* decision that must incentivize *many* choices that politicians make in office.[9]

III. The First Best

We begin by characterizing the voter's optimal selection rule and her first-best payoff. The optimal rule at each date is

$$\pi_i^*(\theta_1, \ldots, \theta_n) = \begin{cases} 1, & \text{if } \theta_i = \max\{\theta_j : j = 1, \ldots, n\}; \\ 0, & \text{otherwise,} \end{cases}$$

for each candidate $i \in C$. Further, the voter prefers that the elected candidate exert maximal effort $a^* = \max A$. The electoral mechanism should therefore aim to give the representative voter a payoff that is as close as possible to

$$v^* \equiv (1-\delta)\,\mathbb{E}_0\left[\sum_{t=1}^{\infty}\sum_{i=1}^{n} \delta^{t-1}\pi_i^*(\theta^t)\int_{\underline{y}}^{\bar{y}} v(y)\,dF(y \mid \theta_i^t, a^*)\right]. \tag{1}$$

Consider first a benchmark setting in which the voter can commit to reward (or punish) candidates with monetary transfers, conditional on their messages $\hat{\theta}$ about their types and (if elected) the realization of the policy output y. Formally, this corresponds to a transfer function

$$r = (r_1, \ldots, r_n) : \Theta \times [\underline{y}, \bar{y}] \to \mathbb{R}^n,$$

such that for each i, and $\hat{\theta}$, the transfer to candidate i, $r_i(\hat{\theta}, y)$, only depends on y if $\pi_i^*(\hat{\theta}) = 1$. We say that the first best is **implementable with transfers** if there exists a transfer function r such that

$$\begin{aligned} &\pi_i^*(\theta_i, \hat{\theta}_{-i})\left[B - c(a^*)\right] + \int_{\underline{y}}^{\bar{y}} r_i(\theta_i, \hat{\theta}_{-i}, y)\,dF(y \mid \theta_i, a^*) \\ &\quad \geq \pi_i^*(\hat{\theta})\left[B - c(a)\right] + \int_{\underline{y}}^{\bar{y}} r_i(\hat{\theta}, y)\,dF(y \mid \theta_i, a), \end{aligned}$$

for all $i \in C$, $\theta_i \in \Theta_i$, $\hat{\theta} \in \Theta$, and $a \in A$. In words: the first best is implementable with transfers if there exists a monetary reward scheme to which the voter can commit—in conjunction with her electoral decision—that induces every candidate to reveal his or her type and choose the maximal effort in office. We emphasize that this constitutes an idealized benchmark: the concerns that motivate this paper are precisely that in real-world contexts, voters cannot commit and have only the blunt instrument of retention or replacement. It is to these contexts that we now turn.

[9] We thank an anonymous referee for highlighting this concern.

IV. Main Result

Recall that our motivating questions concern the necessity of a trade-off between selection and control, whether intrinsic limitations of the electoral mechanism constitute a barrier to electoral accountability, and what precisely constitutes the upper bound of what can be achieved. Our main result provides a positive answer to these questions.

PROPOSITION 1: *Suppose the first best is implementable with transfers. Then, for every* $\epsilon > 0$, *if* δ *is close enough to one, there exists a PBE in which the voter's payoff is within* ϵ *of* v^*.

An alternative statement of Proposition 1 (i.e., the contrapositive) is the following. Suppose that it is *not* possible to approximate the first best in a PBE, when agents are farsighted. Then, it must also not be possible to implement the first best with transfers. In other words, the failure to achieve full accountability is not because of imperfections in the electoral mechanism: the proposition states that it is precisely in these contexts that *commitment and transfers would not help*.

Nonetheless, Proposition 1 leaves open the question of whether there are any contexts in which the first best is indeed implementable with transfers. The answer is *yes*. A special case of our model is the *linear outcome function*, in which the policy outcome y^t is given by a function $Y(\theta_i^t, a_i^t, \varepsilon^t) = \theta_i^t + a_i^t + \varepsilon^t$, where ε^t is an unobservable stochastic shock distributed on $\mathbb{R}$ according to a cumulative distribution function, H, with mean zero. This specification is widely used in the applied literature, including Persson and Tabellini (2002); Ashworth (2005); Gehlbach (2006); Alesina and Tabellini (2007, 2008); and Ashworth and Bueno de Mesquita (2017). The linear outcome function obtains in our model by fixing $F(y \mid \theta_i, a) = H(y - \theta_i - a)$, for all y, θ_i, and a. We can apply standard techniques from the principal-agent literature to demonstrate that the first best is implementable with transfers in this setting. It follows that the first best can be approximated in an equilibrium.

PROPOSITION 2: *Suppose output is generated by the linear outcome function. Then, for every* $\epsilon > 0$, *if* δ *is sufficiently close to one, there exists a PBE in which the voter's payoff is within* ϵ *of* v^*.

V. The Voter's Selection Rule

For the remainder of the paper, we provide intuition for Proposition 1. We first outline the voter's selection rule and give a broad overview of why the selection rule works. We then provide a more detailed explanation of the arguments in the context of a two-candidate example. Proofs of all results are contained in the Appendix.

Step 1 (**Adjusting the Transfers**)**:** We begin with a hypothetical setting in which the voter can supplement her selection decision with a monetary transfer to each candidate. Let $\bar{r} = (\bar{r}_1, \ldots, \bar{r}_n)$ denote a transfer function that implements the first best. We first show that we can "perturb" $\bar{r}$ to another transfer rule r for which the

candidates' incentive constraints hold strictly. That is, under r, each candidate i's relative value from telling the truth θ_i and exerting maximal effort a^* exceeds her value from either lying or shirking by a strictly positive wedge $\beta > 0$:

$$\pi_i^*\left(\theta_i, \hat{\theta}_{-i}\right)\left[B - c\left(a^*\right)\right] + \mathbb{E}_{\tilde{y}}\left[r_i\left(\theta_i, \hat{\theta}_{-i}, \tilde{y}\right) \mid \theta_i, a^*\right] \geq \pi_i^*\left(\hat{\theta}\right)\left[B - c\left(a\right)\right] + \mathbb{E}_{\tilde{y}}\left[r_i\left(\hat{\theta}, \tilde{y}\right) \mid \theta_i, a\right] + \beta \tag{2}$$

whenever $\pi_i^*(\hat{\theta}) \neq \pi_i^*(\theta_i, \hat{\theta}_{-i})$ and/or $a \neq a^*$. Candidates' untruthful reports that leave the electoral outcome unchanged can be ignored as they have no impact on the voter's payoff.

Our goal is to construct an equilibrium of the dynamic game in which a candidate's continuation payoffs—via the voter's decision rule π—simulate the variation in her payoffs in the environment with transfers. Strict incentives allow the voter to preserve candidates' incentives not only with the transfer scheme r but also by way of close approximations of r. This is crucial, because in real-world elections, the voter does not have access to transfers—she can provide incentives only through variation in a candidate's prospect of election. So, the main step is to show that the voter can perturb the optimal selection rule π^* to affect candidates' future payoffs in ways that replicate the incentives arising from transfers.

Step 2 (**Voter's Election Rule**)**:** Under π^*, some candidates are sometimes elected with probability zero or probability one. We next construct a benchmark perturbation of π^* such that all candidates face an interior probability of election, i.e., a probability that lies strictly between zero and one. Let $\hat{\theta}$ denote a profile of messages sent by the candidates; for $\eta \in (0, 1/(2n-1))$, we construct the following perturbation:

$$\pi_i^\eta\left(\hat{\theta}\right) = \begin{cases} \pi_i^*\left(\hat{\theta}\right) - (2n-1)\eta, & \text{if } \pi_i^*\left(\hat{\theta}\right) = 1; \\ \pi_i^*\left(\hat{\theta}\right) + \eta, & \text{otherwise.} \end{cases} \tag{3}$$

Starting from $\pi_i^\eta(\hat{\theta})$, the voter can further adjust each candidate's prospects of election by an additional increment $\phi_i \in (-\eta, \eta)$.[10] The key is that variation in these increments from different announcements and policy outcomes should approximate variation that was achieved when the voter had access to transfers.

We therefore divide time into blocks of T elections, each. At each election in block b, the voter selects each candidate using the perturbed election rule (3) plus an increment $\phi_i^b \in (-\eta, \eta)$ that is constant within the block b and determined at the end of the previous block, i.e., after the policy outcome at date $t = (b-1)T$. The voter determines ϕ_i by way of a *credibility* test and an *eligibility* test. We explain how these tests are implemented, informally, and then describe their significance.

[10] Notice that the low-quality candidate is elected with probability $1 - \sum_{i=1}^n \pi_i^\eta(\hat{\theta}) = n\eta$.

Credibility Test.—At each election within a block, the voter observes a profile of reports, $\hat{\theta}^t$, and generates a profile of candidates' types $\vartheta^t = (\vartheta_1^t, \ldots, \vartheta_n^t)$ that constitutes her "best guess" about each candidate's type at that date.

To formulate this guess, the voter engages in the following introspection: for each candidate i, is her announcement in that election—together with her announcements in previous elections within that block—the messages that I would expect to receive if the candidate were truthful? Specifically: is the empirical transition of each candidate's reported types close to the underlying stochastic process by which her types are generated?

If the answer is *yes*, the voter treats the candidate's announcement as credible, i.e., she sets $\vartheta_i^t = \hat{\theta}_i^t$. If, however, the candidate fails this test, the voter ignores that candidate's announcement and all of her subsequent announcements in that block; instead, she replaces these announcements with simulated messages according to what the voter would have expected under truth telling. In essence, the voter "tunes out" of these untrustworthy candidates.

Eligibility Test.—At the end of the bth block, the voter uses both the profile ϑ^t and the policy outcome y^t at each date $t = (b-1)T + 1, \ldots, bT$ to determine a number:

$$\phi_i^b = \frac{1}{T} \sum_{t=(b-1)T+1}^{bT} r_i\big(\vartheta^t, y^t\big), \tag{4}$$

where we recall that $r_i(\vartheta^t, y^t)$ corresponds to the transfer a candidate received in the static benchmark. The voter then assigns each candidate i an "eligibility" status $e_i^{b+1} \in \{0, 1, \ldots, \lambda T\}$, where $\lambda \in \mathbb{N}$. The eligibility status is determined as follows:

(i) If a candidate's eligibility status at the start of the block b was $e_i^b = 0$, the candidate is assigned $e_i^{b+1} = 0$ if the following test is satisfied:

$$\left| \frac{\phi_i^b}{B - c\big(a^*\big)} \right| < \eta, \tag{5}$$

otherwise the candidate is assigned $e_i^{b+1} = 1$.

(ii) If $e_i^b \in \{1, \ldots, \lambda T - 1\}$, then $e_i^{b+1} = e_i^b + 1$.

(iii) If $e_i^b = \lambda T$, then $e_i^{b+1} = 0$.

The voter then elects each candidate i at each period t in subsequent block $b+1$ with probability

$$\pi_i^b\big(\vartheta^t\big) = \begin{cases} \pi_i^\eta\big(\vartheta^t\big) + \frac{\phi_i^b}{B - c\big(a^*\big)}, & \text{if } e_i^b = 0; \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

A candidate's eligibility status at block b, e_i^b, is interpreted as the number of previous blocks in which the voter has given her a "time-out" by electing her with probability zero at every election within that block. The voter imposes this punishment on any previously eligible candidate that fails the criterion (5). The integer λ is the number of blocks for which a candidate faces this punishment.

The condition defining the eligibility test represents the voter's "budget constraint" in how much variation she can assign to any candidate's prospect of reelection. Recall that, under the perturbed rule (3), the voter can further modify each candidate's prospect of election up or down by η. The eligibility test (5) constrains further variation in the benchmark rule to respect this constraint.

The credibility test was developed by Escobar and Toikka (2013), who used it to construct a mechanism that approximates points in the Pareto frontier of dynamic Bayesian games with independent values. We cannot apply their construction to our model for two main reasons. First, our framework includes moral hazard in addition to adverse selection. In the absence of our novel eligibility test, elected candidates could strategically choose their unobservable actions in future blocks to manipulate their continuation values (or simulated transfers) without being detected. Second, our game is not one of independent values since the voter's utility depends on the elected candidate's type.[11]

VI. Why the Rule Works

We first provide a broad intuition for why the voter's decision rule induces an equilibrium in which her payoff is close to the first-best payoff v^*. We then describe the arguments in more detail in a two-candidate example.

Candidate Incentives.—We want to focus on a candidate's incentives in block b, setting aside the strategic behavior of the remaining candidates as well as the voter's credibility and eligibility tests in that and all subsequent blocks.

To that end, we consider the following thought experiment. At the start of block b, suppose that candidate i expects every other candidate to truthfully report her type and select maximal effort a^* at every election *and* that the candidate i is herself precommitted to truthful reporting and maximal effort in every election starting in the subsequent block $b+1$. Suppose also that the candidate expects to be elected under the first-best rule π^* during elections in block b and at each election from block $b+2$ onward but anticipates being elected under the perturbed rule $\pi_i^b(\hat{\theta}^t)$ defined in (6) at each date $t = bT+1, \ldots, (b+1)T$ in the next block $b+1$. Suppose, finally, that neither the credibility nor eligibility test will be used: the voter believes each candidate's announcement and every candidate remains eligible at the start of the subsequent block.

[11] Note that this difficulty cannot be overcome simply by treating the voter as a mechanism designer, since her first-best payoff does not belong to the candidates' Pareto frontier.

Candidate i's problem is, therefore, to choose a strategy for each election in block b. Observe that the candidate's continuation value at the start of block $b+1$—under our presumption that she remains eligible—is

$$\begin{aligned}
&(1-\delta)\sum_{t=bT+1}^{(b+1)T}\delta^{t-1}\pi_i^b\big(\hat{\theta}_i^t,\theta_{-i}^t\big)\big[B-c\big(a^*\big)\big]+\delta^T\times\text{constant}\\
&\quad=(1-\delta)\sum_{t=bT+1}^{(b+1)T}\delta^{t-1}\Bigg[\underbrace{\pi_i^\eta\big(\hat{\theta}_i^t,\theta_{-i}^t\big)+\phi_i^b\big[B-c\big(a^*\big)\big]}_{\pi_i^b\left(\hat{\theta}_i^t,\theta_{-i}^t\right)}\Bigg]^{-1}\big[B-c\big(a^*\big)\big]\\
&\qquad+\delta^T\times\text{constant}\\
&\quad=(1-\delta)\frac{\sum_{t=(b-1)T+1}^{bT}r_i\big(\hat{\theta}_i^t,\theta_{-i}^t,y^t\big)}{T\big[B-c\big(a^*\big)\big]}\sum_{t=bT+1}^{(b+1)T}\delta^{t-1}\big[B-c\big(a^*\big)\big]+\text{constant}\\
&\quad=(1-\delta)\frac{\sum_{t=(b-1)T+1}^{bT}r_i\big(\hat{\theta}_i^t,\theta_{-i}^t,y^t\big)}{T}\frac{1-\delta^T}{1-\delta}+\text{constant.}
\end{aligned}$$

Since $\lim_{\delta\to 1}(1-\delta^T)/(1-\delta)=T$, it follows that for sufficiently high δ, the candidate's objective function at the start of block b is approximately

$$\begin{aligned}
(7)\qquad &(1-\delta)\sum_{t=(b-1)T+1}^{bT}\delta^{t-1}\pi_i^*\big(\hat{\theta}_i^t,\theta_{-i}^t\big)\big[B-c\big(a_i^t\big)\big]\\
&\quad+(1-\delta)\sum_{t=(b-1)T+1}^{bT}\int_{\underline{y}}^{\bar{y}}r_i\big(\hat{\theta}_i^t,\theta_{-i}^t,y^t\big)\,dF\big(y^t\,|\,\theta_i^t,a_i^t\big)\\
&\approx(1-\delta)\sum_{t=(b-1)T+1}^{bT}\pi_i^*\big(\hat{\theta}_i^t,\theta_{-i}^t\big)\big[B-c\big(a_i^t\big)\big]\\
&\quad+(1-\delta)\sum_{t=(b-1)T+1}^{bT}\int_{\underline{y}}^{\bar{y}}r_i\big(\hat{\theta}_i^t,\theta_{-i}^t,y^t\big)\,dF\big(y^t\,|\,\theta_i^t,a_i^t\big).
\end{aligned}$$

Comparing this objective function with the inequality (2), we observe that if the candidate's objective were *exactly* given by (7), truthfully reporting her type and choosing the optimal action at every date t would be optimal for candidate i.

Recall, however, that the continuation values only *approximate* the incentives under transfers. Nonetheless, our proof shows that an arbitrarily good approximation in the dynamic game of candidate i's incentives in the setting with transfers leads her to choose strategies whose payoff consequences to the voter are arbitrarily

close to the first best, v^*. Moreover, making η arbitrarily small implies that the voter's required distortions of the optimal rule π^* are also arbitrarily small.

Our thought experiment presumed that candidate i will indeed remain eligible in the subsequent block regardless of her opponents' choices. But eligibility in subsequent block $b + 1$ depends on the announcements and actions of *all* the candidates in the previous block b. Since candidate i cannot perfectly anticipate what her opponents will report at each election within the block, she may fear falling afoul of the eligibility test simply because of her opponent's choices. And, subsequent ineligibility means that she foregoes the continuation values that she is supposed to realize in the next block, and our representation of her objective in expression (7) would not apply.

Our proofs therefore establish two facts. First, if a candidate indeed reports truthfully and chooses maximal effort a^* whenever she is elected—or does not deviate "too frequently" from this strategy—she will pass the credibility test and remain eligible with arbitrarily high probability, regardless of her opponents' choices. Second, ineligibility is sufficiently punitive to deter a candidate from any strategy that renders its prospect nontrivial.

Voter Incentives.—The voter subjects an ineligible candidate to λT periods of certain nonelection. This is a severe punishment to the candidate. But it may also severely harm the voter! To see why, recall that at some period of nonelection, candidate i may have the best type, i.e., $\theta_i = \max\{\theta_j : j = 1, \ldots, n\}$. The first-best rule π^* calls on the voter to select this candidate with probability 1. However, the candidate's ineligibility calls on the voter to select the candidate with probability zero. This represents the classic trade-off between selection and control.

This raises two questions. First, how can the voter threaten a candidate with noneligibility if, at one of the subsequent λT elections, the candidate turns out to be the highest quality? Second, to what extent do the voter's costs from implementing the punishment strategy impede her ability to obtain the first-best payoff v^*?

To address the first question, we specify that a "babbling equilibrium" is played at every subsequent election whenever the voter deviates from her election rule: all candidates send uninformative messages to the voter, who elects the best candidate under her prior belief, and whichever candidate is elected chooses the lowest level of effort $a = 0$. Since the voter's payoff from this equilibrium is strictly less than the first-best v^*, the threat of a transition to this equilibrium encourages her to apply the decision rule at each election. This makes her implicit threat to punish a candidate with ineligibility credible.

To address the second question, we show that under the credibility test, so long as a candidate indeed reports her type truthfully and chooses maximal effort—or does not deviate too far from these choices—the prospect that she becomes ineligible is arbitrarily small. This implies that the prospect that the voter is forced to punish a candidate is also arbitrarily small. This, in turn, ensures that the payoff consequences of the trade-off between selection and control that arise whenever the punishment is triggered are arbitrarily small.

A. *The Arguments in More Detail*

To further understand the logic of our result, we develop the arguments in the context of a two-candidate example. Suppose, therefore, that C consists of two candidates ($n = 2$), and each candidate $i = 1,2$ can only be of two types: $\Theta_1 = \{1,3\}$ and $\Theta_2 = \{2,4\}$. In every period, each of candidate i's types is drawn with probability $1/2$, and $y^t = \theta_i^t + a_i^t + \varepsilon^t$, where θ_i^t is the type of the elected candidate, a_i^t is the action chosen from $A = \{0,1\}$, and ε^t is a stochastic shock drawn from the standard normal distribution. Finally, the bad candidate's type is $\theta_0 = 0$, the office rent is $B = 4$, the effort cost is $c(a) = a$, and the voter's period-t payoff is $v(y^t) = y^t$. It is readily checked that the transfer function $r : \{1,3\} \times \{2,4\} \times \mathbb{R} \to \mathbb{R}^2$, defined by

$$r_1\left(\hat{\theta}_1, \hat{\theta}_2, y\right) = \begin{cases} y - \hat{\theta}_2 - \frac{7}{2}, & \text{if } \hat{\theta}_1 > \hat{\theta}_2; \\ \frac{1}{2}, & \text{otherwise,} \end{cases}$$

and

$$r_2\left(\hat{\theta}_1, \hat{\theta}_2, y\right) = \begin{cases} y - \hat{\theta}_1 - 3, & \text{if } \hat{\theta}_1 < \hat{\theta}_2; \\ 1, & \text{otherwise,} \end{cases}$$

implements the first best and that the wedge β defined in equation (2) is equal to one. Note further that $\mathbb{E}[r_1(\theta_1, \theta_2, y)] = \mathbb{E}[r_2(\theta_1, \theta_2, y)] = 0$. The voter's first-best payoff is therefore $v^* = 17/4$. Henceforth, we assume that $\eta \approx 0$ and $\lambda, T \approx \infty$; specifically, we first let $\eta \to 0$, then $\lambda \to \infty$, and then $T \to \infty$. For each candidate $i = 1,2$, let σ_i^* be the strategy that prescribes her to truthfully reveal her type and, if elected, choose the action $a^* = 1$ in every period, regardless of the history of play.

Overview.—Our argument has three key steps. The first step establishes that under the voter's election rule, a candidate i's payoff is bounded from below by a number arbitrarily close to $u_i^* \equiv \mathbb{E}[\pi_i^*(\theta)](B - c(a^*))$ when she chooses σ_i^*. The second step establishes that i's equilibrium payoff is bounded from above by u_i^*. Together, these two steps establish that i's payoff must be arbitrarily close to u_i^* in *any* equilibrium under the voter's election rule. The third and final step shows that only a strategy "close" to σ_i^* secures a payoff arbitrarily close to σ_i^*. This yields the voter a payoff arbitrarily close to the first-best v^*.

Step 1: Suppose that the voter uses the election rule described above. This induces a dynamic game $\Gamma(\eta, \lambda, T, \delta)$ between candidates 1 and 2. We first show that when candidate i plays σ_i^*, her payoff is bounded from below by a number arbitrarily close to $u_i^* \equiv \mathbb{E}[\pi_i^*(\theta)](B - c(a^*))$, regardless of her opponent j's strategy. Let E_i^b denote the event that i passes all eligibility tests in the first $b - 1$ blocks under the strategy

profile (σ_i^*, σ_j). Let $\wp_i^b$ denote the probability of the event E_i^b. Since stage payoffs cannot be negative, i's expected payoff is greater than or equal to

$$\left(1-\delta^T\right)\sum_{b=1}^{\infty}\delta^{(b-1)T}\wp_i^b\,\mathbf{u}_i^b,$$

where for every $b \in \mathbb{N}$,

$$(8)\quad \mathbf{u}_i^b \equiv \mathbb{E}_{\sigma_i^*,\sigma_j}\left[\frac{1-\delta}{1-\delta^T}\sum_{t=(b-1)T+1}^{bT}\delta^{t-(b-1)T-1}\pi_i^b\left(\vartheta^t\right)\times\left(B-c\left(a^*\right)\right)\,|\,E_i^b\right]$$

$$= \mathbb{E}_{\sigma_i^*,\sigma_j}\left[\frac{1-\delta}{1-\delta^T}\sum_{t=(b-1)T+1}^{bT}\delta^{t-(b-1)T-1}\left[\pi_i^\eta\left(\vartheta^t\right)\times\left(B-c\left(a^*\right)\right)+\phi_i^b\right]\,|\,E_i^b\right]$$

$$\approx \mathbb{E}_{\sigma_i^*,\sigma_j}\left[\frac{1}{T}\sum_{t=(b-1)T+1}^{bT}\left[\pi_i^\eta\left(\vartheta^t\right)\times\left(B-c\left(a^*\right)\right)+\phi_i^b\right]\,|\,E_i^b\right]$$

for $\delta \approx 1$. Moreover, the credibility test guarantees that the empirical distribution of the simulated types in every block b converges to the prior distribution of the candidates' types with a probability arbitrarily close to one as $T \to \infty$. Coupled with our observation that $\mathbb{E}[r_1(\theta_1,\theta_2,y)] = 0$, this implies that for sufficiently large T: $\wp_i^b \approx 1$, $\mathbb{E}[\phi_i^b] \approx 0$, and $\mathbf{u}_i^b \approx (B-c(a^*))\mathbb{E}[\pi_i^\eta(\theta)] = 3\mathbb{E}[\pi_i^*(\theta)] - 9\eta \approx u_i^*$.

We conclude that candidate i's payoff is bounded from below by a number arbitrarily close to u_i^* when she plays σ_i^*, regardless of her opponent's strategy.

Parallel arguments show that when the voter uses the prescribed selection rule and when the candidates play σ^*, the voter's payoff approximates the first best. To see why, observe that as long as all candidates remain eligible and pass the credibility test, her average discounted payoff in every block is indeed equal to $v^* = 17/4$. Since $\mathbb{E}[y\,|\,\theta,a] \geq 0$ for all θ and a, her expected payoff is then bounded below by $(1-\delta^T)\sum_{b=1}^{\infty}\delta^{(b-1)T}\wp^b v^*$, where $\wp^b$ denotes the probability that all candidates are eligible and pass the credibility test in every period in block b. By the same logic as above, $\wp^b \approx 1$ in every block b for sufficiently large T, so that the voter's payoff approximates v^*.

Step 2: We showed that a candidate's i's payoff in the dynamic game $\Gamma(\eta,\lambda,T,\delta)$ is bounded from *below* by a number arbitrarily close to u_i^*. We next argue that i's payoff is bounded from *above* by u_i^*. To that end, we construct a variant on $\Gamma(\eta,\lambda,T,\delta)$ in which the eligibility and credibility tests do not apply. In other words: the voter's selection rule is applied to the candidates' reports rather than the output of the credibility test, and adjustments to election probabilities apply regardless of the size of the corresponding transfers. Since this game replicates candidates' incentives under a transfer rule that implements σ^* in dominant strategies, the corresponding sum of

candidate's payoffs $u_1^* + u_2^*$ is an upper bound on the candidates' aggregate payoff in this variant.

Next, we argue that the upper bound on the candidates' aggregate payoff in this variant is also an upper bound on their aggregate payoff in the original game $\Gamma(\eta, \lambda, T, \delta)$. Consider any strategy σ that the candidates play in Γ. Construct a new strategy σ' as follows: σ_i' prescribes the same behavior to candidate i as σ_i in every block where she would be eligible in Γ and prescribes her the same behavior as σ_i^* in every block where she would be ineligible in Γ.

By construction, the difference between the candidates' aggregate payoff from playing σ' in the variant versus the aggregate payoff from playing σ in $\Gamma(\eta, \lambda, T, \delta)$ is only nonzero in blocks where at least one candidate is ineligible. Suppose, for instance, that candidate i becomes ineligible at the start of some block b. That candidate obtains zero for λT blocks in $\Gamma(\eta, \lambda, T, \delta)$. In the modified variant, however, she receives $\mathbb{E}[\phi_i^b \,|\, e_i^b = 1]$ in every period of block b (since candidate i chooses $a = a^*$ whenever elected under σ_i^*), plus u_i^* in each of the next λT blocks. The payoff difference for these blocks is therefore $(1 - \delta^T)\mathbb{E}[\phi_i^b \,|\, e_i^b = 1] + (1 - \delta^{\lambda T})\, u_i^*$. This, divided by $1 - \delta^T$, becomes arbitrarily close to $\mathbb{E}[\phi_i^b \,|\, e_i^b = 1] + \lambda u_i^*$ as $\delta \to 1$. As transfers are bounded above by $3/2$ (in absolute value) and $u_2^* > u_1^* = 3/4$, $\lambda > 2$ ensures that the payoff difference is positive.

Thus, for any strategy σ in the original game $\Gamma(\eta, \lambda, T, \delta)$, the strategy σ' yields the candidates a strictly higher aggregate payoff in the modified game.

Combining Steps 1 and 2: Step 2 showed that $u_1^* + u_2^*$ is an upper bound on the candidates' aggregate payoff in the modified game and that an upper bound in the modified game is itself an upper bound on their aggregate payoff in the original game. This implies that a payoff vector arbitrarily close to (u_1^*, u_2^*) is also arbitrarily close to the Pareto frontier of the set of feasible payoff vectors in the original game. Step 1 further showed that a number arbitrarily close to u_i^* is a lower bound on each candidate i's equilibrium payoff in the original game. We conclude that each candidate i's expected payoff from the game must be arbitrarily close to u_i^* in every equilibrium σ of $\Gamma(\eta, \lambda, T, \delta)$. The same argument yields that i's expected payoff must be close to u_i^* in the continuation game starting at *every* block. This in turn implies that that the probability of ineligibility at block $b + 1$, conditional on eligibility at block b, is close to zero in every equilibrium: $\Pr_\sigma\{e_i^{b+1} \,|\, e_i^b = 0\} \leq (B - u_i^*)(\lambda u_i^*)^{-1} \approx 0$.[12]

Step 3: We show that in any equilibrium of $\Gamma(\eta, \lambda, T, \delta)$, a candidate cannot deviate "too much" from the strategy prescribed by σ_i^*. That is, she must truthfully announce her type and choose $a^* = 1$ if elected, arbitrarily often. To that end, consider an arbitrary equilibrium σ of $\Gamma(\eta, \lambda, T, \delta)$. Let ρ_i^t be the probability

[12] To see this, let $p \equiv \Pr_\sigma\{e_i^{b+1} \,|\, e_i^b\}$: $u_i^* \approx (1 - \delta^T)\, \mathbf{u}_i^b + \delta^T[(1 - p)\, u_i^* + p\, \delta^{\lambda T} u_i^*]$. Rearranging and taking $\delta \approx 1$, we obtain $\mathbf{u}_i^b \approx u_i^* + p\lambda u_i^* < B$, so that $p < (B - u_i^*)(\lambda u_i^*)^{-1}$.

that candidate $i = 1,2$, conditional on being elected in period t, chooses the action $a = 0$. As $u_1^* = 3/4$, it follows from the arguments above that for each block b,

$$\begin{aligned}
\frac{3}{4} &\approx \frac{1-\delta}{1-\delta^T}\,\mathbb{E}_\sigma\left[\sum_{t=1}^{T}\delta^{t-1}\pi_1^*\left(\vartheta^{(b-1)T+t}\right)\left(B - a^{(b-1)T+t}\right)\right] \\
&\approx \frac{1}{T}\,\mathbb{E}_\sigma\left[\sum_{t=1}^{T}\pi_1^*\left(\vartheta^{(b-1)T+t}\right)\left[3\left(1-\rho_1^t\right)+4\rho_1^t\right]\right] \\
&= 3\,\mathbb{E}_\sigma\left[\frac{1}{T}\sum_{t=1}^{T}\pi_1^*\left(\vartheta^{(b-1)T+t}\right)\right] + \mathbb{E}_\sigma\left[\frac{1}{T}\sum_{t=1}^{T}\pi_1^*\left(\vartheta^{(b-1)T+t}\right)\rho_1^t\right] \\
&\approx \frac{3}{4} + \mathbb{E}_\sigma\left[\frac{1}{T}\sum_{t=1}^{T}\pi_1^*\left(\vartheta^{(b-1)T+t}\right)\rho_1^t\right],
\end{aligned}$$

where the last approximation follows from the credibility test, which, for large T, ensures that the empirical distribution of types approximates its invariant distribution with arbitrarily high probability. Hence, the expected proportion of periods in which candidate 1 fails to implement $a^* = 1$ is close to zero. Considering now candidate 1's aggregate payoff, we then have

$$\begin{aligned}
(9)\quad \frac{3}{4} &\approx (1-\delta)\mathbb{E}_\sigma\left[\sum_{b=1}^{\infty}\sum_{t=(b-1)T+1}^{bT}\delta^{t-(b-1)T-1}\left(B - c\left(a^*\right)\right)\left[\pi_1^*\left(\vartheta^t\right) + \frac{\phi_1^b}{B - c\left(a^*\right)}\right]\right] \\
&= \left(1-\delta^T\right)\mathbb{E}_\sigma\left[\sum_{b=1}^{\infty}\delta^{(b-1)T}\left(\frac{1-\delta}{1-\delta^T}\sum_{t=(b-1)T+1}^{bT}\delta^{t-(b-1)T-1}\left(B - c\left(a^*\right)\right)\pi_1^*\left(\vartheta^t\right)\right.\right. \\
&\qquad\qquad \left.\left. + \delta^T\phi_1^{b+1}\right)\right] \\
&\approx \left(1-\delta^T\right)\mathbb{E}_\sigma\left[\sum_{b=1}^{\infty}\delta^{(b-1)T}\left(\frac{1}{T}\sum_{t=(b-1)T+1}^{bT}\left[3\pi_1^*\left(\vartheta^t\right) + r_1\left(\vartheta^t, y^t\right)\right]\right)\right]
\end{aligned}$$

(recall that $\phi_1^1 \equiv 0$). As the transfer rule implements the first best with a wedge β equal to one, the expected frequency of candidate 1's untruthful announcements (affecting the electoral outcome) must be close to zero in the equilibrium σ; otherwise, her payoff would be bounded away from $3/4 = u_1^* = \mathbb{E}[3\pi_1^*(\theta) + r_1(\theta, y)]$. The same logic applies to candidate 2.

Thus, in any equilibrium under the voter's selection rule, the most competent candidate is elected and chooses the voter's ideal action in every period with arbitrarily high probability. Existence of such an equilibrium follows from Fudenberg and Levine (1983). Finally, note that the voter's first-best payoff v^* is strictly larger than her babbling equilibrium payoff of 3. So, for sufficiently large δ, we can rule out profitable deviations for the voter away from the selection rule by imposing that any such deviation is followed by perpetual play of the babbling equilibrium.

B. *Remarks on Discounting*

Our result relies on farsightedness of both voters and politicians. Indeed, Proposition 1 contrasts with a myopic ($\delta = 0$) benchmark.

REMARK 1: *Suppose* $\delta = 0$. *Then, the voter receives her babbling equilibrium payoff,*

$$v^b \equiv \max_{i \in C} \mathbb{E}_{\tilde{\theta}_i^1}\Big[\int_{\underline{y}}^{\bar{y}} v(y)\, dF\big(y \,|\, \tilde{\theta}_i^1, 0\big)\Big],$$

in every PBE.

More generally, for *any* δ it is readily verified that a babbling equilibrium exists and yields the voter's worst equilibrium payoff.[13] However, while we assume that agents have common discount factors for expositional ease, Proposition 1 extends when the voter's and candidates' discount factors belong to an interval $(\bar{\delta}, 1)$, where $\bar{\delta} < 1$. This includes, for example, a setting in which the voter discounts the future more than the candidates.

Alternatively, we could consider a setting in which politicians are farsighted and the voter is wholly myopic, i.e., in which she discounts the future entirely. To understand the consequences of this environment, recall that in our construction, the voter's farsightedness facilitates her commitment to the selection rule defined in Section V. The reason is that any deviation from this rule triggers perpetual play of the babbling equilibrium. When is this punishment least costly to the voter and, thus, her temptation to deviate from the selection rule greatest? Suppose that *all* ordinary candidates (i.e., candidates in C) are ineligible at the start of a block, so that the voter's rule requires her to select the "bad" candidate with type θ_0 at every period within that block. This candidate exerts zero effort whenever elected.

If the voter deviates from this rule by selecting any ineligible candidate, i.e., with type $\theta \in \bigcup_{i \in C} \Theta_i$, she triggers perpetual play of the babbling equilibrium. In particular, she secures zero effort in the period that she deviates and in every future period of the game. However, since the bad candidate's type satisfies $\theta_0 < \min_i(\min \Theta_i)$, the voter nonetheless derives a strict short-term (i.e., within-block) improvement in selection, even under the babbling equilibrium. This temptation is mitigated only when the voter internalizes the long-run cost of zero effort at every future date, which requires sufficient patience. Relatedly, in a setting with pure adverse selection, a myopic voter could not commit to the selection rule. In fact, she receives the babbling equilibrium payoff in every PBE in that setting.

Nonetheless, our environment and construction specialize to important contexts in which a myopic voter facing farsighted politicians *can* achieve the first best.

[13] Standard arguments show that the set of equilibrium payoffs is compact (e.g., Mailath and Samuelson 2006), and thus a best and a worst equilibrium payoff exist.

REMARK 2: *Suppose that* $F(y|\theta_0, 0) = F(y|\theta, 0)$ *for all* $\theta \in \bigcup_{i \in C} \Theta_i$, *and suppose further that the voter is myopic* (*i.e.*, *her discount factor is* $\delta_v = 0$). *Letting* δ *solely denote politicians' common discount factor*, *Proposition 1 extends.*

The condition on output trivially holds in a setting with pure moral hazard. It also applies in our setting with adverse selection and moral hazard when the voter's stage payoff is the product of a candidate's effort and type—as in work discussed above.

To understand why the remark is true, recall that in our construction, the voter's deviation in any period triggers immediate and perpetual play of a zero-effort equilibrium. If $F(y|\theta_0, 0) = F(y|\theta, 0)$ for all $\theta \in \bigcup_{i \in C} \Theta_i$, any two candidates that choose zero effort deliver the same stage payoff to the voter. Thus, the voter receives no immediate gain from abandoning her rule to select an ineligible candidate rather than the bad candidate, and she remains strictly worse off in the long run from deviating. As a consequence, even a myopic voter strictly prefers to stick to the selection rule.

One might conjecture that office rents can be substituted for a high discount factor in a politician's incentives.[14] Because we cannot fully characterize candidates' equilibrium strategies but rather "approximate" them with limiting arguments, we cannot perform comparative statics on primitives such as office rents or the discount factor. Large values of the discount factor imply that average discounted payoffs can be approximated by empirical averages of stage payoffs in every block. This conveys two benefits to the voter. First, because of the credibility test—which ensures that the empirical distribution of each candidate's type is arbitrarily close to the invariant distribution—the voter guarantees every candidate i whose behavior does not deviate too much from σ_i^* a payoff close to u_i^* *regardless of other candidates' actions.* In other words, the selection rule prevents any candidate $j \neq i$ from manipulating the relationship between the voter and candidate i. Second, the perturbations of the winning probabilities approximate sums (as opposed to discounted sums) of the promised simulated transfers in the candidates' expectations of their continuations.[15] While larger office rents facilitate the simulation of promised transfers, they do not help to accomplish these two goals.

VII. Conclusion

We revisit fundamental questions about the limits of political accountability in a model of dynamic elections with adverse selection and moral hazard. We ask: is the trade-off between selection and control fundamental to the electoral mechanism? Are the deficiencies of the electoral mechanism—i.e., the lack of commitment and restriction to binary incentives—to blame for an upper bound on political accountability? What is the upper bound on political accountability?

Whenever the first best can be implemented with transfers and commitment, there exists an equilibrium of the electoral interaction that approximates the first best arbitrarily well, so long as agents are not too impatient. Our result stands in sharp

[14] See, for example, Proposition 5.3 and the preceding discussion in Duggan and Martinelli (2017).
[15] These benefits are highlighted in expressions (8) and (9), respectively.

contrast to existing results in the literature on dynamic elections, which find that voter payoffs are bounded above, regardless of patience. In addition to this possibility result, we explicitly characterize voter behavior in the equilibrium that achieves this payoff. More generally, our paper highlights that electoral institutions need not be a barrier to political accountability. Our contribution is theoretical. Nonetheless, understanding the theoretical limits of accountability is crucial for understanding its practical limitations—in particular, whether these limitations should be attributed to political institutions or, instead, to the agents that interact through them. Our findings therefore highlight the importance of empirical work that explores voter behavior in both experimental (e.g., Landa 2010) and observational contexts (e.g., Aruoba, Drazen, and Vlaicu 2019).

Mathematical Appendix

PROOF OF PROPOSITION 1:

Road Map.—To prove Proposition 1, it suffices to establish that for sufficiently high δ, there exists an equilibrium in which the following happens arbitrarily often: (i) the voter uses a decision rule arbitrarily close to $\pi^*(\cdot)$, (ii) the candidates truthfully report their types, and (iii) the winning candidate belongs to $C \equiv \{1, \ldots, n\}$ and chooses the action a^*.

We begin with two preliminary observations. Firstly, we will focus on equilibria in which candidate 0's probability of being elected only depends on the other candidates' past reports and outputs. It follows that it is optimal for this candidate to choose the lowest action $a = 0$ whenever she is elected. (As this will only occur with an arbitrarily small probability, it will only have a negligible impact on the voter's expected payoff.)

Secondly, it is readily checked that repeating the following one-period "babbling equilibrium" in every period gives a PBE for the infinite horizon model: all candidates send uninformative messages to the voter, who always elects the best candidate according to her prior beliefs, and the elected candidate always chooses the costless action $a = 0$. The voter's expected payoff in this equilibrium is strictly lower than v^*. Therefore, for any candidate equilibrium, attaching the punishment of perpetual play of the babbling equilibrium to out-of-equilibrium voting behavior suffices to ensure (for sufficiently large δ) that such behavior is suboptimal. Although the voter may use mixed strategies, the candidates' messages can be used to generate a public correlating device (Aumann, Maschler, and Stearns 1995), allowing them to detect voter deviations.

Put together, the above two observations obviate the need to consider deviations by candidate 0 and the voter. To obtain an equilibrium giving an expected payoff of v^* to the voter, it suffices to construct a dynamic decision rule (or mechanism) that creates the appropriate incentives for the candidates in C. Under this dynamic rule, equilibrium choices in future periods serve a transfer role, rewarding the candidates for their past actions and revelations of their types. The first step of the proof establishes that we can perturb the transfer function that implements the first-best outcome in such a way that the incentive constraints hold strictly. In the second step

we construct a dynamic decision rule to simulate this transfer rule in our dynamic model, using the candidates' continuation values. Finally, step 3 verifies that the game induced by the dynamic decision rule possesses a PBE in which the voter's payoff is arbitrarily close to v^*.

Step 1 (**Adjusting the Transfer Rule**)**:** Let $\bar{r} = (\bar{r}_1, \ldots, \bar{r}_n)$ be a transfer rule that implements the first-best payoff v^*. Our first step is to perturb $\bar{r}$ to another transfer rule ρ, under which the candidates' incentive constraints hold strictly.

LEMMA 1: *There exist a transfer rule* $\rho = (\rho_1, \ldots, \rho_n)$ *and* $\beta > 0$ *such that the following holds for all* $i \in C$, $\theta_i, \hat{\theta}_i \in \Theta_i$, *and* $\hat{\theta}_{-i} \in \Theta_{-i}$: *if either* $\pi_i^*(\theta_i, \hat{\theta}_{-i}) \neq \pi_i^*(\hat{\theta})$ *or* $\pi_i^*(\hat{\theta}) = 1$ *and* $a \neq a^*$, *then*

$$\pi_i^*\big(\theta_i, \hat{\theta}_{-i}\big)\big[B - c\big(a^*\big)\big] + R_i^*\big(\theta_i, \hat{\theta}_{-i} \,|\, \theta_i\big)$$

$$> \pi_i^*\big(\hat{\theta}_i, \hat{\theta}_{-i}\big)\big[B - c\big(a\big)\big] + R_i\big(\hat{\theta}_i, \hat{\theta}_{-i}, a \,|\, \theta_i\big) + \beta,$$

where $R_i(\hat{\theta}, a \,|\, \theta_i) \equiv \mathbb{E}_{\tilde{y}}[\rho_i(\hat{\theta}, \tilde{y}) \,|\, \theta_i, a]$ *and* $R_i^*(\hat{\theta} \,|\, \theta_i) \equiv R_i(\hat{\theta}, \mathbf{a}(\hat{\theta}_i) \,|\, \theta_i)$.

A proof of Lemma 1 can be found in the online supplementary Appendix. Two observations are in order. First it follows from the lemma that given the transfer rule ρ, every candidate's cost from choosing an action $a \neq a^*$ or misreporting her type (whenever this affects the outcome of the election) is bounded away from zero. Therefore, the candidates' incentive constraints still hold strictly for sufficiently small perturbations of the rules π^* and ρ. Second, the lemma does not rule out deviations from truth telling that leave the electoral outcome unchanged. However, even if one slightly perturbs π^*, such deviations can only have an arbitrarily small impact on the voter's payoff.

Next, let $r_i : \Theta \times \mathbb{R} \to \mathbb{R}$ be defined by

$$r_i\big(\hat{\theta}, y\big) \equiv \rho_i\big(\hat{\theta}, y\big) - \mathbb{E}_\theta\Big[\int_{\underline{y}}^{\bar{y}} \rho_i(\theta, y)\, dF\big(y \,|\, \theta_i, a^*\big)\Big].$$

Here and in what follows, the expectations over type profiles θ are always taken using the invariant distribution p of the Marvov chain governing the evolution of types. Note that the incentive constraints in Lemma 1 still hold if we replace ρ by r.

Step 2 (**The Voter's Dynamic Decision Rule**)**:** It follows from the previous step that the voter's ideal outcome can be supported as an equilibrium outcome if one can appropriately specify continuation play in the dynamic model, so that a deviation by a candidate adversely changes her continuation value. Our next goal is to show that this can be (approximately) achieved if the voter uses an appropriate history-dependent decision rule.

Fix $\eta \in (0, 1/(2n-1))$, and let the decision rule π^{η} be defined by

$$\pi_i^{\eta}\left(\hat{\theta}\right) \equiv \begin{cases} \pi_i^*\left(\hat{\theta}\right) - (2n-1)\eta, & \text{if } \pi_i^*\left(\hat{\theta}\right) = 1; \\ \pi_i^*\left(\hat{\theta}\right) + \eta, & \text{otherwise,} \end{cases}$$

for each $i \in C$. Observe that under this decision rule, candidate 0 is always elected with probability $n\eta$. This allows the voter to reward/punish each candidate $i \in C$ in any period by increasing/decreasing her probability of being elected by some $\varphi_i \in (-\eta, \eta)$.

We define the decision rule as follows. It divides time into blocks of T periods. Each block $b = 1, 2, \ldots$ begins with an "eligibility vector" $e^b = (e_1^b, \ldots, e_n^b) \in \{0, 1, \ldots, \lambda T\}^n$, $\lambda \in \mathbb{N}$, and a vector of promised "rewards" $\varphi^b = (\varphi_1^b, \ldots, \varphi_n^b) \in [-\eta, \eta]^n$ inherited from play in the previous block; we set $e^1 \equiv (0, \ldots, 0)$ and $\varphi^1 \equiv (0, \ldots, 0)$. Then, in each period $t = (b-1)T + 1, \ldots, bT$, events unfold as follows:

(i) The candidates simultaneously report their types.

(ii) Each candidate i's report, $\hat{\theta}_i^t$, is submitted to Escobar and Toikka's (2013) statistical test to assess its credibility, and a profile of types $(\vartheta_1^t, \ldots, \vartheta_n^t)$ is generated as follows: if all of i's previous reports (in block b) passed the test, then $\vartheta_i^t = \hat{\theta}_i^t$; otherwise, ϑ_i^t is obtained from the theoretical distribution of i's types.

(iii) The voter selects each candidate $i \in C$ with probability $\pi_i^b(\vartheta^t) \equiv \pi_i^{\eta}(\vartheta^t) + \varphi_i^b$ if $e_i^b = 0$ (i.e., if candidate i is "eligible") and with probability 0 otherwise.

(iv) The elected candidate, say j, chooses an action a^t.

(v) The outcome y^t is observed by all agents.

At the end of the bth block, each candidate i's eligibility and reward for the next block are determined as follows. First, we calculate

$$\phi_i^b \equiv \frac{1}{T} \sum_{t=(b-1)T+1}^{bT} r_i\left(\vartheta^t, y^t\right);$$

then

$$e_i^{b+1} \equiv \begin{cases} 0 & \text{if either } e_i^b = \lambda T, \text{ or } e_i^b = 0 \text{ and } \left|\left[B - c\left(a^*\right)\right]^{-1} \phi_i^b\right| < \eta; \\ e_i^b + 1, & \text{otherwise;} \end{cases}$$

and then

$$\varphi_i^{b+1} \equiv \begin{cases} \left[B - c\left(a^*\right)\right]^{-1} \phi_i^b, & \text{if } e_i^b = e_i^{b+1} = 0; \\ 0, & \text{otherwise.} \end{cases}$$

Step 3 (**The Equilibrium**)**:** The dynamic decision rule defined above induces a dynamic game between the n candidates in C, which we denote by $\Gamma(\eta, \lambda, T, \delta)$. To complete the proof of the theorem, it remains to establish that for appropriately chosen parameters η, λ, T, and δ, this game has a PBE in which the voter's expected payoff is approximately v^*.

For each $i \in C$, let σ_i^* be the strategy that prescribes candidate i to truthfully reveal her type when campaigning and choose action a^* when in office, regardless of the previous history of play. It is readily checked that for every strategy profile σ (and every T), the average discounted payoff to i in each block $b \in \mathbb{N}$ can be approximated by

$$\mathbb{E}_\sigma\left[\mathbf{1}_{\{e_i^b=0\}}\,\frac{1}{T}\sum_{t=1}^{T}\left[\pi_i^\eta\left(\vartheta^{(b-1)T+t}\right)+\varphi_i^b\right]\left[B-c\left(a_i^{(b-1)T+t}\right)\right]\right]$$

by letting $\delta \to 1$. Now suppose that candidate i plays σ_i^* and, therefore, chooses action a^* whenever she is elected in block $b-1$. It follows from Lemma 5.1 in Escobar and Toikka (2013) that the empirical distribution of the sequence of report profiles generated by the test in each block converges to the invariant distribution p as $T \to \infty$, irrespective of the candidates' actual reports. By definition of r_i, the expected values of ϕ_i^{b-1} and φ_i^b (conditional on $e_i^{b-1} = 0$) approach 0 if we let $T \to \infty$. This has the following two implications for i's payoff. First, i's expected block b payoff, conditional on being eligible in block b, is approximately $\mathbb{E}_\theta[\pi_i^\eta(\theta)[B - c(a^*)]]$, regardless of the other candidates' strategies. Second, i's probability of remaining eligible approaches 1 as we let $T \to \infty$, so that for every λ, we can choose T large enough that i is eligible arbitrarily frequently. It follows that candidate i can guarantee herself a payoff arbitrarily close to $\mathbb{E}_\theta[\pi_i^\eta(\theta)[B - c(a^*)]]$ by playing σ_i^*, and if each candidate $i \in C$ plays σ_i^*, then the voter receives an expected payoff arbitrarily close to $\mathbb{E}_{\theta,y}[\sum_{i=1}^n \pi_i^\eta(\theta)v(y) \mid a^*]$. This is formally stated in the next lemma, whose complete proof can be found in the online supplementary Appendix.

LEMMA 2: *For all $\eta > 0$ and $\lambda \in \mathbb{N}$, the following holds for game $\Gamma(\eta, \lambda, T, \delta)$ if we let $T \to \infty$ and then $\delta \to 1$:*

(*i*) *for each $i \in C$, the payoff to candidate i from playing σ_i^* approaches $\mathbb{E}_\theta[\pi_i^\eta(\theta)[B - c(a^*)]]$, irrespective of the other candidates' strategies, and*

(*ii*) *if each candidate $i \in C$ plays σ_i^*, then the voter's expected payoff approaches $\mathbb{E}_{\theta,y}[\sum_{i=1}^n \pi_i^\eta(\theta)v(y) \mid a^*]$.*

Let $\eta \approx 0$. It follows from the second part of Lemma 2 that for any $\varepsilon > 0$ and sufficiently large T and δ, the voter's expected payoff is within ε of v^* if every candidate i plays σ_i^*. Therefore, our final task is to show that $\Gamma(\eta, \lambda, T, \delta)$ has a PBE in which every candidate i's strategy prescribes the same choices as σ_i^* arbitrarily frequently in every block. Our next result draws on Lemma 2(i) to show that this is the case for sufficiently large λ, T, and δ.

LEMMA 3: *For any $\gamma \in (0,1)$, if we let $\eta \to 0$, then $\lambda \to \infty$, then $T \to \infty$, and finally $\delta \to 1$, there exists a PBE for $\Gamma(\eta, \lambda, T, \delta)$ in which the expected empirical frequency of deviations from behavior prescribed by $(\sigma_1^*, \ldots, \sigma_n^*)$ is less than γ.*

We provide a complete proof of Lemma 3 in the online supplementary Appendix and outline here the main steps of the argument. First, one can show that if we let $\eta \to 0$, then $\lambda \to \infty$, then $T \to \infty$, and finally $\delta \to 1$, the strategy profile $(\sigma_1^*, \ldots, \sigma_n^*)$ generates a payoff vector that is arbitrarily close to the utilitarian optimum and, therefore, to the Pareto frontier of the set of feasible payoffs in $\Gamma(\eta, \lambda, T, \delta)$. Coupled with Lemma 2(i), this implies that each candidate i's equilibrium payoff from any block where she is eligible must be arbitrarily close to $\mathbb{E}_\theta[\pi_i^\eta(\theta)[B - c(a^*)]]$. We can then use Lemma 1 to show that if candidate i deviated too often from σ_i^* in an equilibrium, then her expected payoff would be bounded away from $\mathbb{E}_\theta[\pi_i^\eta(\theta)[B - c(a^*)]]$. It follows that in any PBE of $\Gamma(\eta, \lambda, T, \delta)$, each candidate i must play in accordance with σ_i^* in an arbitrarily large proportion of blocks with an arbitrarily high probability. Finally, Fudenberg and Levine's (1983) existence theorem guarantees that such an equilibrium exists, thus completing the proof of Lemma 3. ∎

PROOF OF PROPOSITION 2:

Applying techniques from the principal-agent literature (e.g., Laffont and Tirole 1993), we can construct a transfer function r that implements the first best. The corollary then follows directly from Proposition 1. Specifically, define first the function $\mathbf{a}: \mathbb{R} \to \mathbb{R}_+$ as any C^1 function that satisfies the following conditions: $\mathbf{a}(x) = a^*$ and $\mathbf{a}'(x) > 0$ for all $x \in \bigcup_{i \in C} \Theta_i$. Then, for each $i \in C$, let $r_i: \Theta \times [\underline{y}, \overline{y}] \to \mathbb{R}$ be defined by

$$r_i(\hat{\theta}, y) \equiv \pi_i^*(\hat{\theta})\Big[c'\big(\mathbf{a}(\hat{\theta}_i)\big)\big[y - \hat{\theta}_i - \mathbf{a}(\hat{\theta}_i)\big] + c\big(\mathbf{a}(\hat{\theta}_i)\big) + \int_{m_i(\hat{\theta})}^{\hat{\theta}_i} c'\big(\mathbf{a}(z)\big)\, dz - B\Big],$$

where $m_i(\hat{\theta}) \equiv \max_{j \neq i} \hat{\theta}_j$. Fix $i \in C$ and $\hat{\theta}_{-i} \in \Theta_{-i}$. To prove the proposition, we must show that given the decision rule π^* and the transfer rule r_i, candidate i is always better off truthfully reporting her true type θ_i and choosing the optimal action a^* than deviating to some untruthful report and/or to some action $a \neq a^*$.

We begin by showing that whenever it wins the election after making some report $\hat{\theta}_i \in \Theta_i$, candidate i is better off choosing action $\mathbf{a}(\hat{\theta}_i) = a^*$. Given the transfer rule $r_i(\cdot)$, an optimal action for i must be a solution to

$$\max_{a \in A} R_i\big(\hat{\theta}_i, \hat{\theta}_{-i}, a \,|\, \theta_i\big) - c(a),$$

or, equivalently,

$$\max_{a \in A} c'\big(\mathbf{a}(\hat{\theta}_i)\big)(\theta_i + a) - c(a).$$

Assuming that the feasible set is $\mathbb{R}_+$ (instead of A), it follows from the first-order condition (and the strict concavity of the objective function) of this problem that its

unique solution is $\mathbf{a}(\hat{\theta}_i)$. As $\mathbf{a}(\hat{\theta}_i) = a^* \in A$, we conclude that it is also the unique solution to the above maximization problem.

We now turn to candidate i's optimal report. We must consider two cases: first, where candidate i wins the election by truthfully reporting her type θ_i and, second, where she loses the election by truthfully reporting her type. We begin with the first case, i.e., where $\pi_i^*(\theta_i, \hat{\theta}_{-i}) = 1$. Anticipating that she will choose the optimal action if elected, candidate i's expected payoff from making any report $\hat{\theta}_i > m_i(\hat{\theta})$ is given by $B - c(\mathbf{a}(\hat{\theta}_i)) + R_i(\hat{\theta}_i, \hat{\theta}_{-i} | \theta_i)$. Maximizing this function with respect to $\hat{\theta}_i$ in $\mathbb{R}$ boils down to solving

$$\max_{\hat{\theta}_i \in \mathbb{R}} c'\big(\mathbf{a}\big(\hat{\theta}_i\big)\big)\big(\theta_i - \hat{\theta}_i\big) + \int_{m_i(\hat{\theta})}^{\hat{\theta}_i} c'\big(\mathbf{a}\big(z\big)\big)\, dz.$$

The only solution to the first- and second-order conditions in Θ_i is $\hat{\theta}_i = \theta_i$. Moreover, making a report $\hat{\theta}_i < m_i(\hat{\theta})$ (so that $\pi_i^*(\hat{\theta}_i, \hat{\theta}_{-i}) = 0$) gives a payoff of 0, whereas reporting her true type θ_i yields

$$\Big[B - c\big(\mathbf{a}\big(\theta_i\big)\big)\Big] + R_i\big(\theta_i, \hat{\theta}_{-i} | \theta_i\big) = \int_{m_i(\theta_i, \hat{\theta}_{-i})}^{\theta_i} c'\big(\mathbf{a}\big(z\big)\big)\, dz > 0$$

(recall that $c' > 0$). Hence, candidate i is better off truthfully reporting her type.

We now turn to the second case, where $\pi_i^*(\theta_i, \hat{\theta}_{-i}) = 0$, so that $\theta_i < m_i(\theta_i, \hat{\theta}_{-i})$. Consider a deviation from θ_i to any report $\hat{\theta}_i$ such that $\pi_i^*(\hat{\theta}_i, \hat{\theta}_{-i}) = 1$ (so that $\hat{\theta}_i > m_i(\hat{\theta}) > \theta_i$). The benefit of such a deviation is equal to

$$\begin{aligned} & c'\big(\mathbf{a}\big(\hat{\theta}_i\big)\big)\big(\theta_i - \hat{\theta}_i\big) + \int_{m_i(\hat{\theta})}^{\hat{\theta}_i} c'\big(\mathbf{a}\big(z\big)\big)\, dz \\ & \quad \leq c'\big(\mathbf{a}\big(\hat{\theta}_i\big)\big)\big(\theta_i - \hat{\theta}_i\big) + \Big[\hat{\theta}_i - m_i\big(\hat{\theta}\big)\Big] c'\big(\mathbf{a}\big(\hat{\theta}_i\big)\big) \\ & \quad = -c'\big(\mathbf{a}\big(\hat{\theta}_i\big)\big)\Big[m_i\big(\hat{\theta}\big) - \theta_i\Big] < 0, \end{aligned}$$

where the inequality follows from the fact that $c'(\mathbf{a}(\cdot))$ is a strictly increasing function. This completes the proof of Proposition 1. ∎

PROOF OF REMARK 1:

Suppose $\delta = 0$, so that the players' expected payoffs only depend on the first period's electoral outcome. Trivially, the voter never elects candidate 0 in a PBE, as she could otherwise increase her expected payoff by replacing this "bad" candidate with any other candidate in C. In addition, it follows immediately from sequential rationality that the elected candidate always chooses the costless action $a = 0$ in equilibrium. Hence, the objective of every candidate in stage 2 is only to maximize her probability of winning the election, which in turn implies that all the types of any given candidate i must have the same probability of winning in equilibrium.

Take an arbitrary PBE. For each candidate $i \in C$, let $\{\Theta_i^1, \ldots, \Theta_i^{K_i}\}$, with $1 \leq K_i \leq |\Theta_i|$, be the partition of Θ_i such that all types in partition element Θ_i^k make the same announcement $\hat{\theta}_i^k$ in stage 2, which differs from the announcements made by the types in the other partition elements. (Thus, this equilibrium is pooling if $K_i = 1$ for all i and is separating if $K_i = |\Theta_i|$ for all i.) Let

$V(\Theta_i')$ be the voter's expected payoff when she elects some candidate i conditional on the latter's type belonging to some subset $\Theta_i' \subseteq \Theta_i$, i.e.,

$$V(\Theta_i') \equiv \mathbb{E}_{\tilde{\theta}_i^1}\Big[\int_{\underline{y}}^{\bar{y}} v(y)\, dF\big(y \,|\, \tilde{\theta}_i^1, 0\big) \,|\, \tilde{\theta}_i^1 \in \Theta_i'\Big].$$

Suppose first that in the equilibrium under consideration, some candidate i wins the election with probability 1, regardless of the realization of the type profile $(\theta_1, \ldots, \theta_n)$; so that

$$\min_{k \in K_i} V\big(\Theta_i^k\big) \geq \max_{j \neq i} \max_{\ell \in K_j} V\big(\Theta_j^\ell\big).$$

The voter's expected payoff is then $V(\Theta_i)$, and we consequently have

$$V(\Theta_i) \geq \min_{k=1,\ldots,K_i} V\big(\Theta_i^k\big) \geq \max_{j \neq i} \max_{\ell=1,\ldots,K_j} V\big(\Theta_j^\ell\big) \geq \max_{j \neq i} V(\Theta_j),$$

where the first and last inequalities follow from the fact that for each $j \in C$, $V(\Theta_j)$ is a convex combination of the $V(\Theta_j^\ell)$ terms. But since $V(\Theta_i) \geq \max_{j \neq i} V(\Theta_j)$, the voter's expected payoff must also be equal to $V(\Theta_i)$ in the babbling equilibrium.

Suppose now that depending on the realization of the type profile $(\theta_1, \ldots, \theta_n)$, at least two candidates in C may win the election with positive probability. If candidate i wins with positive probability when she announces $\hat{\theta}_i^k$, then $V(\Theta_i^k) \geq \max_{\ell'} V(\Theta_j^{\ell'})$ for all $j \neq i$; otherwise, candidate j could increase her (ex ante) probability of winning by always making the announcement $\hat{\theta}_j^{\ell'}$ that maximizes $V(\Theta_j^{\ell'})$. Similarly, if candidate j wins with positive probability when she announces $\hat{\theta}_j^\ell$, then $V(\Theta_j^\ell) \geq \max_{k'} V(\Theta_i^{k'})$ for all $i \neq j$. It follows that there exists a constant $\bar{V}$ such that for any two potential winners i and j, we have $V(\Theta_i^k) = V(\Theta_j^\ell) = \bar{V}$ for all $k = 1, \ldots, K_i$ and $\ell = 1, \ldots, K_j$. Hence, the voter's equilibrium payoff is $\bar{V}$. By the same logic as above, we must have

$$\bar{V} = V(\Theta_i) \geq \max_{j \in C} \max_{\ell=1,\ldots,K_j} V\big(\Theta_j^\ell\big)$$

for any potential winner i. It follows that the voter's payoff in the babbling equilibrium must also be equal to $\bar{V}$; otherwise, she could profitably deviate from that equilibrium by always selecting i. ∎

REFERENCES

Acharya, Avidit, Elliot Lipnowski, and João Ramos. 2021. "Optimal Political Career Dynamics." http://elliotlipnowski.com/wp-content/uploads/accountability-ALR.pdf.

Alesina, Alberto, and Guido Tabellini. 2007. "Bureaucrats or Politicians? Part I: A Single Policy Task." *American Economic Review* 97 (1): 169–79.

Alesina, Alberto, and Guido Tabellini. 2008. "Bureaucrats or Politicians? Part II: Multiple Policy Tasks." *Journal of Public Economics* 92 (3–4): 426–47.

Alt, James, Ethan Bueno de Mesquita, and Shanna Rose. 2011. "Disentangling Accountability and Competence in Elections: Evidence from U.S. Term Limits." *Journal of Politics* 73 (1): 171–86.

Aruoba, S. Borağan, Allan Drazen, and Razvan Vlaicu. 2019. "A Structural Model of Electoral Accountability." *International Economic Review* 60 (2): 517–45.

Ashworth, Scott. 2005. "Reputational Dynamics and Political Careers." *Journal of Law, Economics, and Organization* 21 (2): 441–66.

Ashworth, Scott, and Ethan Bueno de Mesquita. 2006. "Delivering the Goods: Legislative Particularism in Different Electoral and Institutional Settings." *Journal of Politics* 68 (1): 168–79.

Ashworth, Scott, and Ethan Bueno de Mesquita. 2017. "Unified versus Divided Political Authority." *Journal of Politics* 79 (4): 1372–85.

Aumann, Robert J., Michael Maschler, and Richard E. Stearns. 1995. *Repeated Games with Incomplete Information*. Cambridge, MA: MIT Press.

Banks, Jeffrey S., and Rangarajan K. Sundaram. 1993. "Adverse Selection and Moral Hazard in a Repeated Elections Model." In *Political Economy: Institutions, Information, Competition, and Representation*, edited by William A. Barnett, Melvin J. Hinich, and Norman J. Schofield, 295–311. Cambridge, UK: Cambridge University Press.

Banks, Jeffrey S., and Rangarajan K. Sundaram. 1998. "Optimal Retention in Agency Problems." *Journal of Economic Theory* 82 (2): 293–323.

Bernhardt, Dan, Odilon Câmara, and Francesco Squintani. 2011. "Competence and Ideology." *Review of Economic Studies* 78 (2): 487–522.

Besley, Timothy. 2007. *Principled Agents?: The Political Economy of Good Government*. Oxford, UK: Oxford University Press.

Camargo, Braz, and Arianna Degan. 2019. "Accountability and Political Competition." https://economie.esg.uqam.ca/wp-content/uploads/sites/54/2019/01/2019-01_docdt_eco.compressed.pdf.

Canes-Wrone, Brandice, Michael C. Herron, and Kenneth W. Shotts. 2001. "Leadership and Pandering: A Theory of Executive Policymaking." *American Journal of Political Science* 45 (3): 532–50.

Di Tella, Rafael, and Julio J. Rotemberg. 2018. "Populism and the Return of the 'Paranoid Style': Some Evidence and a Simple Model of Demand for Incompetence as Insurance against Elite Betrayal." *Journal of Comparative Economics* 46 (4): 988–1005.

Duggan, John. 2014. "A Folk Theorem for Repeated Elections with Adverse Selection." *Political Science Research and Methods* 2 (2): 213–42.

Duggan, John. 2017. "Term Limits and Bounds on Policy Responsiveness in Dynamic Elections." *Journal of Economic Theory* 170: 426–63.

Duggan, John, and César Martinelli. 2017. "The Political Economy of Dynamic Elections: Accountability, Commitment, and Responsiveness." *Journal of Economic Literature* 55 (3): 916–84.

Duggan, John, and César Martinelli. 2020. "Electoral Accountability and Responsive Democracy." *Economic Journal* 130 (627): 675–715.

Escobar, Juan F., and Juuso Toikka. 2013. "Efficiency in Games with Markovian Private Information." *Econometrica* 81 (5): 1887–1934.

Fearon, James D. 1999. "Electoral Accountability and the Control of Politicians: Selecting Good Types versus Sanctioning Poor Performance." In *Democracy, Accountability, and Representation*, edited by Adam Przeworski, Susan C. Stokes, and Bernard Manin, 55–97. Cambridge, UK: Cambridge University Press.

Ferejohn, John. 1986. "Incumbent Performance and Electoral Control." *Public Choice* 50 (1/3): 5–25.

Fouirnaies, Alexander, and Andrew B. Hall. 2018. "How Do Electoral Incentives Affect Legislator Behavior?" http://www.andrewbenjaminhall.com/Fouirnaies_Hall_Electoral_Incentives.pdf.

Fudenberg, Drew, and David Levine. 1983. "Subgame-Perfect Equilibria of Finite- and Infinite-Horizon Games." *Journal of Economic Theory* 31 (2): 251–68.

Gailmard, Sean. 2010. "Accountability and Principal-Agent Theory." In *The Oxford Handbook of Public Accountability*, edited by Mark Bovens, Robert E. Goodin, and Thomas Schillemans, 90–105. Oxford, UK: Oxford University Press.

Gehlbach, Scott. 2006. "Electoral Institutions and the National Provision of Local Public Goods." *Quarterly Journal of Political Science* 2: 5–25.

Gordon, Sanford C., Gregory A. Huber, and Dimitri Landa. 2007. "Challenger Entry and Voter Learning." *American Political Science Review* 101 (2): 303–20.

Ingham, Sean. 2019. *Rule by Multiple Majorities: A New Theory of Popular Control*. Cambridge, UK: Cambridge University Press.

Kartik, Navin, and Richard Van Weelden. 2019. "Reputation Effects and Incumbency (Dis)Advantage." *Quarterly Journal of Political Science* 14 (2): 131–57.

Laffont, Jean-Jacques, and Jean Tirole. 1993. *A Theory of Incentives in Procurement and Regulation*. Cambridge, MA: MIT Press.

Landa, Dimitri. 2010. "Selection Incentives and Accountability Traps: A Laboratory Experiment." https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1640033.

Landa, Dimitri, and Patrick Le Bihan. 2018. "Does Unbundling Policy Authority Improve Accountability?" *Journal of Politics* 80 (3): 933–47.

Manin, Bernard, Adam Przeworski, and Susan C. Stokes. 1999. "Elections and Representation." In *Democracy, Accountability, and Representation,* edited by Adam Przeworski, Susan C. Stoke, and Bernard Manin, 29–54. Cambridge, UK: Cambridge University Press.

Persson, Torsten, and Guido Tabellini. 2002. *Political Economics*. Cambridge, MA: MIT Press.

Roche, MaryClare. 2020. "Non-consecutive Executives: Changing Preferences in a Changing World." http://www.maryclareroche.com/uploads/1/1/6/2/116207021/non-consecutive_executives.pdf.

Van Weelden, Richard. 2013. "Candidates, Credibility, and Re-election Incentives." *Review of Economic Studies* 80 (4): 1622–51.

Zudenkova, Galina. 2018. "Electoral Accountability and Interdistrict Competition." *American Economic Journal: Microeconomics* 10 (3): 143–76.